ALEYAH DAWKINS

Department of Mathematical Sciences
Carnegie Mellon University
Pittsburgh, Pennsylvania 15213 USA

Email: adawkins@andrew.cmu.edu

EDUCATION EXPERIENCE

NSF MPS-Ascend Postdoctoral Research Fellow, Carnegie Mellon University — August 2024–Present
Scientific Mentor: Michael Young

Ph.D. in Mathematics, George Mason University (GMU) — August 2019–May 2024
Advisors: Geir Agnarsson and Rachel Kirsch

B.S. in Mathematics, James Madison University — August 2012–May 2016

INDUSTRY EXPERIENCE

Los Alamos National Laboratory, Los Alamos, New Mexico — August 2023–August 2024
Guest Student

U.S. Census Bureau, Suitland, Maryland — December 2016–August 2019
Statistician

RESEARCH INTERESTS

Combinatorics extremal graph theory, graph labeling, Ramsey theory, design theory

PUBLICATIONS AND PREPRINTS

1. A. Dawkins, A. Nappo, J. Tong, and C. Zimbizi. *Wild number for spanning trees in complete and complete bipartite graphs* arXiv:2610.22142, Submitted, 2026.
2. A. Dawkins and R. Kirsch. *Ore plus Turán* arXiv:2310.11452, Submitted, 2026.
3. A. Dawkins and R. Kirsch. *Maximizing the number of cliques in Kr+1-free graphs with forbidden properties* arXiv:2609.25196, Submitted, 2026.
4. A. Cao, A. Dawkins, J. Hutchins, and O. Luce *Radio gracefulness of Moore graphs and beyond*, Australasian Journal of Combinatorics, 2026. arXiv:2510.00228
5. M. Bayer, S. Borgwardt, T. Chambers, S. Daugherty, A. Dawkins, D. Deligeorgaki, H. Liao, T. McAllister, A. Morrison, G. Nelson, and A. Vindas-Meléndez *Combinatorics of generalized parking-function polytopes*, Discrete & Computational Geometry, 2025. [arXiv:2403.07387]
6. A. Dawkins, V. Gupta, M. Kempton, W. Linz, J. Quail, H. Richman, Z. Stier. *A Ricci flow on graphs from effective resistance*, Journal of Combinatorics, 2025. [arXiv:2403.01151]
7. A. Dawkins, V. Gupta, M. Kempton, W. Linz, J. Quail, H. Richman, Z. Stier. *Node resistance curvature in Cartesian graph products*, Journal of Combinatorics, 2025. [arXiv:2403.01037]
8. K. Isham, L. Monroe, K. Lakhotia, A. Dawkins, D. Hwang, and .A. Kubicek, *Edge-Disjoint Spanning Trees on Star-Product Networks*, 2025 IEEE International Parallel and Distributed Processing Symposium (IPDPS). [arXiv:2403.12231]
9. G. Agnarsson, E. Alhajjar, and A. Dawkins. *On locally finite ordered rooted trees and their rooted subtrees*, Discrete Applied Mathematics, 2025. [arXiv:2312.11379]
10. A. Dawkins, J. Elder, P. E. Harris, K. Harry, D. Smith, and B. E. Tenner. *On the Lucky and Displacement Statistics of Stirling Permutations*, Journal of Integer Sequences, 2024. [arXiv:2403.03280]

RESEARCH EXPERIENCE

Graduate Research Assistant — Los Alamos National Laboratory
Advisor: Dr. Laura Monroe — June 2023–August 2023

Topic: *Constructing edge-disjoint spanning trees in post-exascale networks*

Graduate Research Assistant — George Mason University
Advisor: Dr. Rachel Kirsch — August 2022–May 2023

Title: *Edge conditions for Hamiltonicity and related properties in K_{r+1}-free graphs*

Graduate Research Assistant — George Mason University
Industrial Immersion Program (IIP) — June 2022–August 2022

Advisors: Dr. Geir Agnarsson (GMU Math) and Dr. Elie Alhajjar (West Point Math)
Title: *On locally finite ordered rooted trees and their rooted subtrees*

Graduate Assistant Mentor — George Mason University
Mason Experimental Geometry Lab (MEGL) — January 2022–May 2022

Advisor: Dr. Ben Schweinhart and Dr. Erika Roldan
Title: *Random 3D Polyforms*

Graduate Research Assistant — George Mason University
Industrial Immersion Program (IIP) — May 2021–August 2021

Advisors: Dr. Maria Emelianenko (GMU Math), Dr. Ming Tian (GMU Physics), and Dr. Fred Wieland (Mosaic ATM)
Title: *Quantum algorithms for Unmanned Air Vehicles (UAV) trajectory optimization*

Graduate Research Assistant — George Mason University
NSF Improving Undergraduate STEM Education — August 2020–May 2021

Advisors: Dr. Robert Sachs and Dr. Jill Nelson

MENTORING

Undergraduate Research Advisor — Carnegie Mellon University

Summer Undergraduate Applied Mathematics Institute (3 students) — Summer 2026
Title: *Wild Edges in Edge-Colored Graphs*

Summer Undergraduate Applied Mathematics Institute (3 students) — Summer 2025
Title: *Radio Labelings and the Moore bound*

Graduate Assistant Mentor — George Mason University
Mason Experimental Geometry Lab (2 students) — January 2020–May 2020

Advisor: Dr. Neil Epstein
Title: *Content formulas in algebras derived from Grassmanians*

TEACHING EXPERIENCE

Instructor of Record — Carnegie Mellon University

21-102 - Exploring Modern Mathematics — Spring 2027
21-102 - Exploring Modern Mathematics — Spring 2026

Instructor of Summer Academy — Carnegie Mellon University

Modern Mathematics (20 high school students) — Summer 2026
Introduction to Graph Theory (26 high school students) — Summer 2024
Exploring the Worlds of Geometry (10 high school students) — Summer 2024

Tartans Scholar Tutor — Carnegie Mellon University

Various classes: Calculus, Linear Algebra — Fall 2024 - Spring 2025

Instructor of Record — George Mason University

MATH 114 - Analytic Geometry and Calculus II	Summer 2022
MATH 114 - Analytic Geometry and Calculus II (2 courses)	Summer 2021
MATH 114 - Analytic Geometry and Calculus II	Summer 2020
MATH 108 - Introductory Calculus with Business Applications	Summer 2020

Graduate Learning Assistant — George Mason University

PhD Launchpad Program — Summer 2022

Graduate Teaching Assistant — George Mason University

Math Tutoring Center	January 2022–May 2022
MATH 113 - Analytic Geometry and Calculus I	August 2021–December 2021
MATH 114 - Analytic Geometry and Calculus II	January 2021–May 2021
MATH 114 - Analytic Geometry and Calculus II	August 2020–December 2020
MATH 113 - Analytic Geometry and Calculus I	January 2020–May 2020
MATH 113 - Analytic Geometry and Calculus I	August 2019–December 2019

FUNDING AND AWARDS

- National Science Foundation, Mathematical and Physical Sciences Ascending Postdoctoral Research Fellowship, PI: Aleyah Dawkins ($300,000), 2024-27
- Dean's Graduate Award for Excellence in Research, GMU, 2023-24 ($1250)
- Doctoral Research Scholars Assistantship, Office of Provost, GMU, 2023-24 ($22,000)
- Graduate Research Excellence Award, Department of Mathematics, GMU, 2023 ($1000)
- Dean's Award for Excellence in Service, GMU, 2022 ($1250)
  - for contributions to improve Mason Experimental Geometry Lab outreach to the community
- TC Lim Award for Excellence in Teaching, Department of Mathematics, GMU, 2022 ($1000)
- TC Lim Award for Excellence in Teaching, Department of Mathematics, GMU, 2021 ($1000)

PRESENTATIONS

Invited Talks

- JMM special session on Extremal Combinatorics and Beyond, *Ramsey numbers with wild edges*, January 2027
- JMM special session on Recent Developments in Graph Labeling Problems, *Radio Graceful Labelings of Generalized Petersen and Moore Graphs*, January 2027
- JMM NAM Haynes-Granville-Browne Session, *Wild number for spanning trees in complete and bipartite graphs*, January 2027
- 57th Southeastern International Conference on Combinatorics, Graph Theory & Computing, March 2026
- Kansas Women in Math Symposium, *Graph homomorphisms and radio graceful graphs*, April 2025
- Iowa State Discrete Math Seminar, *Radio Graceful Labelings of Graphs*, March 2025
- OURFA^{2}M^2, Crash Course on *Drawing Connections Through Graphs*, February 2025
- JMM special session on Graphs, Hypergraphs, and Extremal Combinatorics, *Hamiltonicity and related properties in K_{r+1}-free graphs*, January 2025
- University of Massachusetts - Amherst Discrete Math Seminar, *Hamiltonicity and related properties in K_{r+1}-free graphs*, May 2024

- AMS Spring Eastern Sectional Meeting special session on Fresh Researchers in Algebra, Combinatorics, and Topology (FRACTals), *Edge conditions for Hamiltonicity in K_{r+1}-free graphs*, April 2024
- JMM special session on Applications of Extremal Graph Theory to Network Design, *Constructing Spanning Trees in Post-Exascale Networks*, January 2024
- JMM special session on Extremal and Probabilistic Combinatorics, *Hamiltonicity and related properties in K_{r+1}-free graphs*, January 2024
- AMS Fall Southeastern Sectional Meeting special session on Extremal and Probabilistic Combinatorics, *Post-Exascale Network Design*, October 2023
- Graduate Online Combinatorics Colloquium, *Hamiltonicity and related properties in K_{r+1}-free graphs*, September 2023
- AMS Spring Southeastern Sectional Meeting special session on Recent trends in Structural and Extremal Graph Theory, *Edge conditions for Hamiltonicity and related properties in K_{r+1}-free graphs*, March 2023

Contributed Talks and Posters

- ACO Seminar, *Ramsey numbers with wild edges*, Carnegie Mellon University, December 2026
- GSP Seminar, *The radio number of Moore graphs, cages, and random graphs*, Carnegie Mellon University, November 2025
- NAM MATHfest, *Hamiltonicity and related properties in K_{r+1}-free graphs*, September 2023
- ACORN Workshop Poster Session, *Edge conditions for Hamiltonicity and related properties in K_{r+1}-free graphs*, March 2023
- Howard Mathematics Graduate Student Seminar, *Edge conditions for Hamiltonicity in K_{r+1}-free graphs*, February 2023
- XVIII Escuela de Verano en Matemáticas Discretas, *Edge conditions for Hamiltonicity in K_{r+1}-free graphs*, January 2023
- IIP Summer Symposium, *Cyber security models and attacks*, August 2022
- BRIDGES Workshop, *Cheeger constant of Cartesian products of Cayley graphs arising from generalized dihedral groups*, June 2022
- JMM, *Cheeger constant of Cartesian products of Cayley graphs arising from generalized dihedral groups*, April 2022
- IIP Summer Symposium, *Quantum algorithms for Unmanned Air Vehicles (UAV) trajectory optimization*, August 2021
- MEGL Symposium, *Content formulas in algebras derived from Grassmanians* (jointly with undergraduate students), May 2020

ORGANIZING

- AMS Fall Eastern Sectional Meeting 2025, Co-organizer for special session on "Combinatorics and Graph Theory: The Audience Counts"
- AMS Fall Western Sectional Meeting 2024, Co-organizer for special session on "Gender Equity in the Mathematical Sciences (GEMS) of Combinatorics"
- JMM 2024, Co-organizer for *AMS Special Session on Ricci Curvatures of Graphs and Applications to Data Science*

CONFERENCES AND WORKSHOPS

- BIRS Bridging Theoretical Discrete Math Advancements with Open Source Resource Development 2026
- Combinatorics at the Confluence 2026
- NAM MATHfest 2025
- PIMS-BIRS-UBCO GEMS of Optimization 2025
- AIM workshop on Graph Theory: structural properties, labelings, and connections to applications 2024

- Early Career Researchers in Combinatorics (ECRiC24) 2024 (hosted by International Centre for Mathematical Sciences)
- Graduate Research Workshop in Combinatorics (GRWC) 2023
- AMS Mathematics Research Communities (MRC) conference 2023: Ricci Curvatures of Graphs and Applications to Data Science
- Extremal Graphs arising from Designs and Configurations (hosted by Banff International Reserach Station)
- GEMS of Combinatorics 2023 (hosted by AIM)
- CIME: Mentoring for Equity 2023 (hosted by MSRI/SLMath)
- Modern Math Workshop 2022 (organized by MSRI/SLMath)
- MSRI - Random Graphs 2022 Summer Graduate School
- Building Relationships for an Inclusive and Diverse Group of Emerging Students (BRIDGES) Workshop 2022 (University of Utah)

PROFESSIONAL SERVICE AND OUTREACH

- CMU Math Club Colloquium on *Ramsey Theory: Generalizations and Variations*, September 2026
- Women Who Lead 2026, Pittsburgh Public Schools
- Pittsburgh Community Science Workshop
  - Volunteer: Fall 2024, Spring 2025
- Panel on "Graduate School", MATHfest 2023
- Student Research Talks (StReeTs) Seminar (GMU)
  - Co-organizer: Fall 2021, Spring 2022, Fall 2022, Spring 2023
- Society for Industrial and Applied Mathematics (GMU Chapter)
  - Vice President: 2021-2023
- Mason Experimental Geometry Lab
  - Lab Co-manager: Fall 2021, Spring 2022
  - Graduate Outreach Assistant: Fall 2021, Spring 2022
  - Graduate Assistant Mentor: Spring 2020, Spring 2022
- Graduate and Professional Student Association (GAPSA) (GMU)
  - Mathematics Representative: 2021-2022
- Mathematics Outreach
  - Volunteer at East-John Youth Center: Summer 2021